

**Smart Fisher Lanka is an AI-based decision support system  
designed for Sri Lankan fishermen**

Project ID: 25-26J-177

Project Proposal Report

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B.Sc. (Hons) Degree in Information Technology specialized in  
Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology  
Sri Lanka

August 2025

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Research Component: AI-Driven Fish Quality Grading and Weight  
Estimation Module

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## DECLARATION

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The above candidate is carrying out research for the undergraduate Dissertation under my supervision.

## Abstract

The Sri Lankan fishery industry is gravely threatened in maintaining consistent fish quality levels, with an economic implication and loss of market competitiveness. Hand grading of conventional fish is subjective, time-consuming, and plagued by human error, and its effect primarily falls on the small-scale fishermen and exporters who have no means of access to sophisticated quality assessment equipment.

This project envisions the deployment of a lightweight, real-time AI-driven fish quality grading system based on EfficientNet-B0 convolutional neural network architecture, optimized for high accuracy and efficient mobile deployment. The system will classify fish freshness, damage extents, and general quality grades from image analysis of mobile phone camera-captured photos immediately after harvest. Utilizing transfer learning techniques with MobileNetV2 and deploying the trained model to TensorFlow Lite, the solution will possess offline functions critical to maritime environments with limited connectivity.

The system suggested completes the principal void of available, impartial fish quality estimation tools designed for Sri Lankan species. Unlike existing global solutions, the system will be trained on local species datasets and customized for fishermen's and exporters' applications with varying technical proficiency. The solution will also contain empirical volume modeling to derive fish weight from density measurement, providing additional value for business use.

Expected outcomes are improved standardization of fish quality, minimized post-harvest losses, improved market prices for quality fish, and increased competitiveness of Sri Lankan fish exports in foreign markets. The mobile-based character of the system ensures access and real-time decision-making ability for stakeholders in the fishing value chain.

**Keywords:** Deep Learning, Fish Quality Grading, Weight Estimation, Length-Weight Relationship, Image Processing, Sri Lankan Fisheries, Real-Time Inference, Computer Vision, Mobile AI

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# 1. INTRODUCTION

## 1.1 Background and Literature survey

Sri Lanka's fisheries industry plays a vital role in ensuring national food security, supporting rural livelihoods, and contributing to economic development. The sector includes marine, inland, and aquaculture resources that provide essential protein and micronutrients to the population while also generating employment across a wide socio-economic network. In addition to its domestic importance, the industry contributes significantly to export earnings, making quality assurance a critical factor in maintaining international competitiveness. National reports and industry analyses highlight the importance of adopting modern technologies and data-driven approaches to improve efficiency, sustainability, and market transparency within the fisheries sector [1], [2].

Despite its significance, the industry continues to face major challenges related to post-harvest management and quality control. Studies and national reports indicate that post-harvest losses can reach up to 40% of the total catch in certain fisheries, particularly in multi-day operations [11], [12]. These losses are primarily caused by improper handling, delays in transportation, and the absence of standardized grading mechanisms. Currently, fish quality assessment is largely performed through manual sensory inspection, where graders evaluate characteristics such as gill color, eye clarity, skin texture, and odor. Although widely practiced, this method is inherently subjective and highly dependent on individual expertise, environmental conditions, and human factors such as fatigue. As a result, it often leads to inconsistent grading, pricing disputes, and inefficiencies in the market [2], [3].

To address similar challenges globally, recent research has explored the application of artificial intelligence and computer vision techniques in fish classification and quality assessment. In particular, deep learning approaches based on Convolutional Neural Networks (CNNs) have demonstrated strong performance in image-based recognition tasks. Early implementations often relied on lightweight architectures such as MobileNet, which are designed for efficiency and low computational cost, making them suitable for mobile and embedded systems [4], [7]. These models utilize techniques such as depthwise separable convolutions to reduce processing requirements while maintaining acceptable levels of accuracy.

More recent advancements in deep learning have introduced improved architectures such as EfficientNet, which achieve higher accuracy through compound scaling of network depth, width, and resolution. Compared to earlier lightweight models, EfficientNet provides better feature extraction capabilities while maintaining computational efficiency, making it highly suitable for real-time applications in resource-constrained environments. In addition, transfer learning techniques—where

models are initialized using weights trained on large-scale datasets such as ImageNet—have been widely adopted to improve generalization and reduce the need for extensive labeled datasets in domain-specific applications [3], [5].

However, a significant limitation identified in the existing literature is the lack of context-specific solutions tailored to tropical fisheries such as those in Sri Lanka. Many AI-based fish classification and grading systems are developed using datasets of temperate species, such as salmon and cod, captured under controlled laboratory conditions [6], [8], [13]. These models often fail to generalize effectively to real-world environments where lighting conditions, background complexity, fish orientation, and species diversity vary significantly. As a result, their applicability in local fisheries remains limited.

Furthermore, many existing solutions remain at the experimental or prototype stage, lacking full system integration for practical deployment. Several approaches rely on cloud-based processing, expensive imaging equipment, or controlled environments, making them unsuitable for small-scale fishermen operating in rural or offshore conditions [9]. This creates a gap between research innovations and real-world usability, particularly for stakeholders who require simple, cost-effective, and accessible tools.

Another key limitation is that most current systems focus on single-task outputs, such as species identification or freshness detection, without providing a comprehensive quality assessment framework. There is limited research on integrating multiple functionalities—including grading, damage detection, and weight estimation—into a unified system that supports real-time decision-making. This reduces the practical value of existing approaches in commercial fisheries operations.

In addition to technological limitations, external factors such as climate change, resource depletion, and regulatory challenges further increase the need for reliable and data-driven solutions within the fisheries sector [14], [15]. These challenges highlight the importance of developing systems that not only improve accuracy but also enhance transparency, traceability, and sustainability.

Therefore, this research aims to bridge the gap between theoretical advancements in deep learning and practical deployment by developing an AI-driven fish quality grading and weight estimation system tailored to Sri Lankan fisheries. The proposed solution leverages modern deep learning architectures and mobile technologies to provide an efficient, scalable, and context-aware system capable of operating in real-world conditions. By addressing both technical and practical limitations identified in the literature, this study contributes toward improving quality standardization, reducing post-harvest losses, and supporting the long-term sustainability of the fisheries industry

## 1.2 Research Gap

Although the fisheries sector is a major contributor to Sri Lanka's economy, employing over half a million people and supporting national food security, it continues to face several persistent and interconnected technological, operational, and environmental challenges that hinder its modernization and long-term sustainability [10].

One of the most critical issues is the high level of post-harvest loss. A significant proportion of the marine fish catch—estimated to be up to 40% in multi-day fisheries—experiences quality deterioration during handling, storage, and transportation [11]. These losses are primarily attributed to inadequate preservation methods, delays in processing, and the absence of rapid and objective grading mechanisms at the point of first sale. As a result, the industry suffers from reduced economic returns, decreased export quality, and inefficiencies in resource utilization [12].

Another major gap lies in the lack of context-specific digital and AI-based solutions. Most existing research and technological implementations for fish quality assessment are developed using datasets from temperate regions, focusing on species such as salmon and cod [13]. These models do not generalize effectively to tropical species found in Sri Lanka, such as skipjack tuna (Kelawalla), yellowfin tuna (Thora), and other commercially important fish. Differences in morphology, environmental conditions, and spoilage patterns lead to inaccurate predictions when such models are applied in local contexts.

Furthermore, many advanced fish grading systems proposed in the literature are not practically accessible to the majority of stakeholders. These systems often require expensive hardware, controlled imaging environments, or continuous connectivity, making them unsuitable for small-scale fishermen and local market vendors who operate under resource-constrained conditions. This creates a significant gap between technological advancements and real-world applicability.

In addition, existing approaches are typically limited to single-task functionalities, such as species classification or freshness detection, without providing a comprehensive quality assessment framework. There is a lack of integrated systems capable of performing multi-parameter analysis, including species identification, quality grading, damage detection, and weight estimation within a unified workflow. This limitation reduces the practical usefulness of such systems in supporting real-time decision-making in fisheries operations.

Moreover, there is limited evidence of solutions that have been validated under real-world Sri Lankan conditions, such as fish landing sites, open markets, and varying environmental settings. Many studies are conducted under controlled laboratory environments, which do not accurately reflect the complexities and variability of real operational scenarios [3]. This highlights a gap in translating research outcomes into deployable and reliable systems for field use.

Finally, broader environmental and regulatory challenges, including climate change impacts, resource depletion, and illegal, unreported, and unregulated (IUU) fishing, further emphasize the need for reliable, data-driven tools to support transparency and sustainability in the fisheries sector [14], [15]. The absence of integrated, intelligent systems limits the ability to monitor quality, ensure compliance, and maintain competitiveness in both local and international markets.

Therefore, there is a clear need for a comprehensive, context-aware, and accessible solution that leverages modern artificial intelligence techniques to address these limitations. This research aims to bridge these gaps by developing an AI-driven fish quality grading and weight estimation system tailored specifically to the Sri Lankan fisheries context, integrating multiple functionalities into a single practical and deployable platform.

### **1.3 Research Problem**

The Sri Lankan fisheries industry greatly needs an integrated, objective, and cost-effective technology for grading fish quality, exactly fitted to the multi-faceted situation of tropical small-scale fisheries.

Reliance on subjective, manual appraisal leads to inconsistent quality at sale, repeated price lacks of agreement and ingrained information asymmetry that hurts small-scale fishers the most. This perpetuates a system of missed incomes, livelihoods laid bare, and low market transparency.

Existing automated tools are out of reach: They require significant initial investment, infrastructure, or connectivity unavailable outside of big cities or landing sites. Their failure to adapt to local species, unpredictable environmental conditions, and existing workflow guarantees that most AI grading tools remain unused or unusable by legitimate stakeholders the very ones most exposed to economic, climatic, and regulatory stress.

The research challenge is thus as much technical and practical as social designing durable multi-label grading of tropical fish with mobile AI as it is solving how to get to a system that is embedded in fisher workflows from catch to sale, applies across a variety of species and ecosystems, and is offline.

The solution will have to deliver not only laboratory accuracy, but working empowerment of Sri Lankan fishermen and market actors, supporting evidence-based decision-making, waste minimization, ensuring fair market prices, and facilitating sustainable growth of industry.

## 2. OBJECTIVES

### 2.1 Main Objectives

The primary objective of this project is to design, develop, and deploy a lightweight, real-time, mobile-enabled fish quality grading and weight estimation system tailored for the Sri Lankan fisheries sector. The system leverages the EfficientNet-B0 deep learning architecture, which provides a balance between high classification accuracy and computational efficiency, making it suitable for mobile-based deployment.

Using only a smartphone camera, the system enables automatic and objective assessment of fish species, freshness level, physical damage, and estimated weight. The solution is specifically designed for key commercial Sri Lankan fish species, including:

- Skipjack Tuna (Balaya / Kelawalla)
- Indian Scad (Linna)

This system addresses major industry challenges such as inconsistent manual grading, lack of standardized pricing, and post-harvest quality losses. By replacing subjective human judgment with AI-based classification, it improves transparency, fairness, and efficiency in fish trading processes.

The application is optimized for real-world fisheries environments, including landing sites, coastal markets, and fishing vessels. It is designed to function efficiently even under limited connectivity conditions, ensuring usability in rural and offshore contexts.

By providing standardized and confidence-based grading outputs, the system supports fishermen, traders, and exporters in making informed decisions regarding pricing, storage, and distribution. This contributes to economic empowerment, reduced waste, and improved sustainability within the fisheries supply chain.

Ultimately, this project goes beyond technological innovation and aims to support digital transformation in the Sri Lankan fisheries industry through practical, accessible, and intelligent decision-support tools.

## **2.2. Specific Objectives**

- To develop a deep learning-based fish classification model using EfficientNet-B0, trained on Sri Lankan fish species such as Skipjack Tuna (Balaya) and Indian Scad (Linna) for accurate species identification, freshness grading, and damage detection.
- To design and implement a user-friendly mobile application that enables real-time fish quality assessment using smartphone-based image capture.
- To integrate an image-based weight estimation module using morphometric analysis and species-specific length–weight relationships.
- To evaluate and validate the model performance against expert human graders in real-world environments such as fish markets and landing sites.
- To ensure the system is scalable, cost-effective, and adaptable for future expansion to additional fish species within the Sri Lankan fisheries domain..

### **2.2.1. User Requirements**

- The app must function flawlessly on low-cost, readily available Android phones.
- Fishers, suppliers, and retailers must have the ability to photograph an item and get instant real-time detailed grading (species, freshness, visible defects, estimated weight) with no technical expertise or additional hardware.
- The app must function in coastal, rural, or market environments, with multiple language support (Sinhala, Tamil, English) and uncomplicated step-by-step instructions.
- The system needs to generate confidence scores, explicit explanations of grading outcomes, and facilitate record keeping or convenient reporting to meet regulatory or export requirements.

### **3. METHODOLOGY**

The project methodology is structured to create systematically, train, deploy, and validate a mobile AI-powered fish quality grading system for optimal performance in Sri Lankan fisheries.

#### **3.1 Key Stages of the Process**

##### **3.1.1 Data annotation and collection**

In this research, the data collection and annotation phase was a critical step in building a robust dataset for training the proposed fish quality grading system. The image dataset was prepared using real-world fish samples captured under practical fisheries environments in Sri Lanka, including landing sites, local markets, and coastal areas.

Images were captured using multiple smartphone devices to ensure variability in camera quality, resolution, and environmental conditions. The devices used for data collection include:

- iPhone 11 Pro
- iPhone 7 Plus
- Samsung Galaxy M11

Using multiple devices helped introduce natural diversity in image quality, lighting response, and sensor characteristics, improving the generalization ability of the trained models.

During data acquisition, fish specimens were photographed from different angles, distances, and lighting conditions to simulate real operational scenarios. Both whole-body and partial views of fish were included to enhance model robustness.

After collection, all images were carefully annotated and labeled according to the system requirements. Annotation was performed based on three main tasks:

1. Fish detection (fish / non-fish classification)
2. Species identification (e.g., Skipjack Tuna, Indian Scad, GrayMullet ,etc.)
3. Quality grading labels (Grade A, B, C for selected species)

Annotation was carried out manually using visual inspection criteria commonly applied in fisheries, ensuring accuracy in labeling freshness, physical damage, and overall quality condition.

To improve model performance and reduce overfitting, data augmentation techniques were applied during preprocessing. These techniques included:

- Image flipping (horizontal and vertical)
- Random cropping and resizing
- Rotation at different angles
- Brightness and contrast adjustment
- Slight zoom-in and zoom-out transformations

These augmentation methods helped simulate real-world variations such as fish orientation changes, handling differences, and varying environmental lighting conditions commonly observed in fish markets.

The combination of multi-device image capture, manual annotation, and augmentation ensured the creation of a diverse and high-quality dataset suitable for training deep learning models such as EfficientNet-B0, enabling accurate fish detection, species classification, and quality grading under real-world conditions.



*Figure 3.1.1: skip jack tuna (Grade -B)*



*Figure 3.1.1: skip jack tuna (Grade -C)*



*Figure 3.1.1: S skip jack tuna (Grade -A)*

### **3.1.2 Preprocessing and augmentation**



- After data collection, the images underwent a systematic preprocessing stage to ensure consistency and improve the quality of input data for the deep learning model. Since the dataset was collected from real-world environments using different mobile devices and under varying lighting conditions, preprocessing was essential to standardize the images before training.
- Initially, all images were resized to a fixed input dimension suitable for the EfficientNet-B0 architecture, ensuring uniformity across the dataset. This step helps the model process images efficiently without being affected by varying original resolutions. In addition, pixel values were normalized to a standard range to stabilize the learning process and improve convergence during training.
- Noise reduction techniques were also applied to minimize background disturbances commonly found in real-world fish market environments, such as shadows, water reflections, and cluttered surfaces. This helped the model focus more on relevant fish features such as body shape, skin texture, and eye clarity.
- To improve the model's ability to generalize to unseen data, data augmentation techniques were applied during training. These techniques artificially expanded the dataset by introducing controlled variations to the existing images. The following augmentation methods were used:
  1. Horizontal flipping to simulate different fish orientations
  2. Random cropping to focus on different regions of the fish body
  3. Rotation at various angles to handle non-uniform image capture positions
  4. Brightness and contrast adjustment to represent different lighting conditions
  5. Zoom in/out transformations to simulate varying camera distances
- These augmentation strategies are particularly important in fisheries environments, where images are rarely captured under controlled conditions.
- By combining preprocessing and augmentation, the dataset was transformed into a more diverse and robust training set. This significantly improved the model's ability to accurately perform fish detection, species classification, and quality grading under real-world conditions, even when image quality varies due to environmental factors or device differences.
- 

### **3.1.3 Building and training the model**

- **Base Model:** EfficientNet-B0, a lightweight convolutional neural network deployable on mobile platforms, will be employed as the base.
- **Transfer Learning:** Employing pre-trained weights from the ImageNet dataset will accelerate training on the specific fish dataset.
- **Multi-Label Classification:** The model will be trained to perform several tasks simultaneously damage detection, and freshness grading.

- **Weight Estimation Regression:** A second regression will estimate fish weight based on image data normalized against morphometric data specific to the species.

### 3.1.4 Weight estimation methodology

- Users will take pictures of fish with a calibrated known size object (e.g., ruler, reference marker) in close proximity to the fish to create scale and enable accurate conversion of measurement.
- Sophisticated image processing techniques are employed by the system to segment the fish from the background and recognize anatomical landmarks (i.e., the snout and tail fork) to accurately gauge the length of fish.
- Pixel width and length measurements are converted to real-world dimensions through the use of the calibration reference for consistent measurement regardless of position or distance from the camera.
- Weight of fish is initially estimated from the generally accepted empirical length-weight relation for the species provided by:

$$W = a * L^b$$

K is a species-specific constant usually ranging from 700 to 900.

- For greater precision, the system estimates girth—the circumference around the thickest part of the fish's body—by estimating width in images and calculating girth as approximately twice the width. This allows for the use of the length-girth empirical formula:

$$\text{Weight} = (\text{Length} * \text{Girth}^2)/2$$

where W is weight, L is length, and a, b are species-specific constants derived from regionally calibrated biological data.

- To get more precise estimates, the model incorporates morphometric information by approximating body width at a set of standardized points along the fish's body to estimate body volume. Volume is calculated by summing elliptical cross-sectional area multiplied by segment lengths.

- Estimated volume can be made even more accurate by multiplying estimated volume by mean species-specific tissue density to capture variation in body condition that might not be detected by length alone.
- The system integrates machine learning regression models and paired image-measurement-ground truth training data to enhance and optimize traditional formula outcomes, adapting dynamically to species morphology and environmental factors.
- The multi-parameter approach is optimized for computational efficiency for mobile deployment as well as scientific precision, enabling feasible use in Sri Lankan fisheries with research-grade reliability.

### **3.1.5 Model deployment and optimization**

- The model will be compressed and optimized (quantization, pruning) for on-device, real-time inference.
- It will be built to TensorFlow Lite and distributed in a Flutter mobile application executable on Android and iOS operating systems.
- Offline capability will be prioritized to enable grading feature in connectivity-low situations.

### **3.1.6 Mobile application development**

- The user interface shall guide the user through image capture, grading outcome interpretation, and export or cloud syncing optionally.
- Multi-language interface support (Sinhala, Tamil, English) and user-friendliness will make the application accessible to fishers, traders, and exporters of all skill levels.

### **3.1.7 Validation and field testing**

- Model output will be compared with expert human grader judgment at landing sites and market sites.
- User experience, environmental robustness, and performance metrics will drive ongoing improvement.
- Validation trials will incorporate a range of fish species and realistic usage conditions to validate the system's operational effectiveness.

## **3.2 Tools and Technology**

The project employs deep learning platforms such as TensorFlow and Keras to make AI models, with TensorFlow Lite enabling fast real-time inference on mobile

platforms without connectivity to the cloud. The mobile application is coded using Flutter, offering flexibility and robust performance on resource-limited smartphones with cross-platform support on Android and iOS. The basic model architecture is MobileNetV2, a convolution neural network that performs well in terms of both high accuracy and low computational needs, and this makes offline inference within feasible areas in remote fisheries environments.

To prepare datasets, annotation tools like Labellmg and CVAT are used to create accurate bounding boxes and multi-label annotations on fish images, which are a requirement for efficient model training. Data augmentation is performed using Keras' ImageDataGenerator, which simulates various visual transformations to enhance model resilience under various field and image conditions. Backend services via Node.js or Python Flask APIs can optionally be deployed for data storage and analytics across the network when remote network access is available. Development environment includes Python 3.x with Jupyter notebooks and TensorBoard for developing and training the model and visualizing the training, and Android Studio for developing and testing the mobile application for stability and performance on devices.



*Figure 3.3: Technology Stack*

### 3.3. Overall System Overview

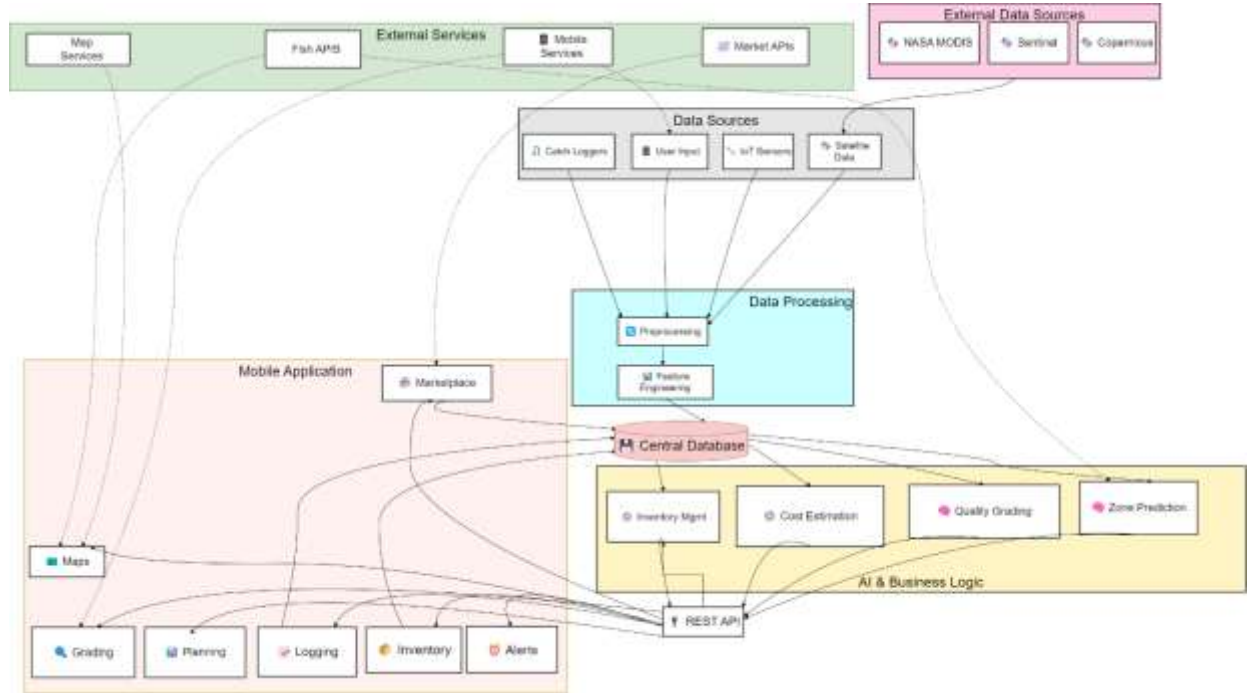


Figure 3.4: Overall System Overview Diagram

Our proposed system is based on an integrated, holistic architecture to address the multidimensional problems of Sri Lankan fishermen. The infrastructure is designed in distinct layers that facilitate seamless data flow from collection to decision-making information. Externally sensed satellite data from NASA MODIS, Sentinel provides underlying oceanographic parameters (SST and chlorophyll concentrations), which feed into our processing data layer alongside inputs from IoT sensors, catch loggers, and manual user input. This data undergoes rigorous preprocessing and feature engineering before being deposited in a centralized database. Business logic and AI layer is the brain of our system, comprising bespoke machine learning models for fish zone prediction (Random Forest/SVM/Neural Networks-based) and fish quality grading (Mobile Net CNN architecture with transfer learning-based), as well as business rules for dynamic trip cost calculation and inventory management. These components communicate through a RESTful API gateway that exposes processed intelligence to a mobile application, enabling features like interactive map-based zone visualization, AI-powered quality grading, trip planning functionality, digital catch logging, marketplace integration, inventory management, and license expiration alerts. The architecture engages in two-way communication with external services like mapping APIs (Google Maps/MapBox), market data feeds, mobile device services, and specialty fisheries APIs, creating an integrated decision-support ecosystem that transforms raw environmental and operational data into actionable intelligence for fishermen at both individual and commercial scales

### 3.4. Component overview

The fish quality grading system is modular in architecture to enable every functional unit to integrate in a seamless manner, hence providing an integrated solution perfectly suited for application on mobile devices. The process starts from data collection and annotation stage, where images of fish over calibration objects are gathered from landing sites and fish markets. The images are thoroughly annotated for size, physical damage, freshness, and species, thereby creating a detailed dataset crucial in enabling proper training of models.

Following data collection, preprocessing and data augmentation are conducted on the images. Normalization of image size, luminance, and contrast is done, as well as segmentation of the fish from backgrounds for visual representation. Rotation, flipping, and color augmentation are done to boost the dataset diversity and give the model the ability to be resilient to real-world variation in surroundings.

The prepared images are then analyzed by the feature extraction module to detect major anatomical features like the snout, tail, and body's widest points. Pixel measurements are calibrated to real world units using the calibration object as reference to enable precise morphometric data extraction necessary for accurate weight estimation and grading.

The core of the system is a number of AI models that are trained to do specific tasks: species and freshness classification, damage detection, and estimation of weight through regression analysis. The models make use of the annotated data and extracted features to make accurate appraisals for each fish specimen.

To attain best performance on handheld, the trained models quantized and pruned are transformed to TensorFlow Lite. This provides real-time efficient inference that can work offline and can be used to function within fisheries environments with poor connectivity.

The mobile application is the user interface, guiding the users through the image acquisition process, demonstrating object calibration inclusion, and showing the grading results in an easy-to-use multi-language user interface. It also supports optional features like data export and cloud synchronization.

Finally, the system also comprises a validation and feedback loop where AI predictions are continually compared with expert human graders and physical measurements. This process relies on user feedback and environmental metadata to continually refine and enhance the accuracy and reliability of the system, rendering the solution dynamic and responsive to field conditions.

Hierarchical, component-based design leads to the system being scalable, maintainable, and flexible, having been specifically made to address the challenges of Sri Lankan fisheries while prioritizing practical useability and scientific excellence.

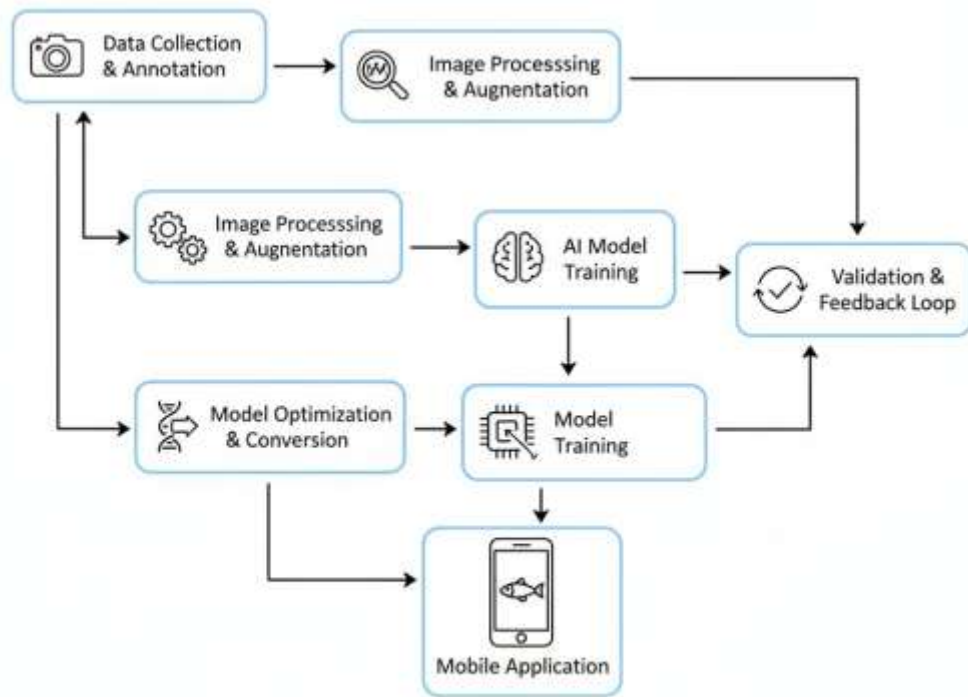


Figure 3.5: Component diagram

## 4. GANTT CHART

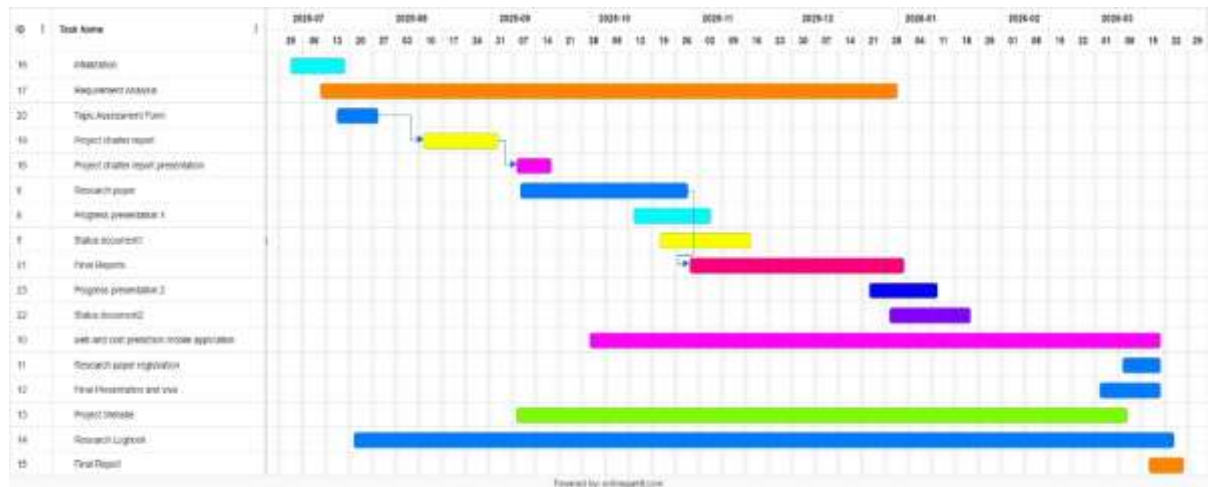


Figure 4.1: Gantt chart



## 5. WORK BREAKDOWN STRUCTURE

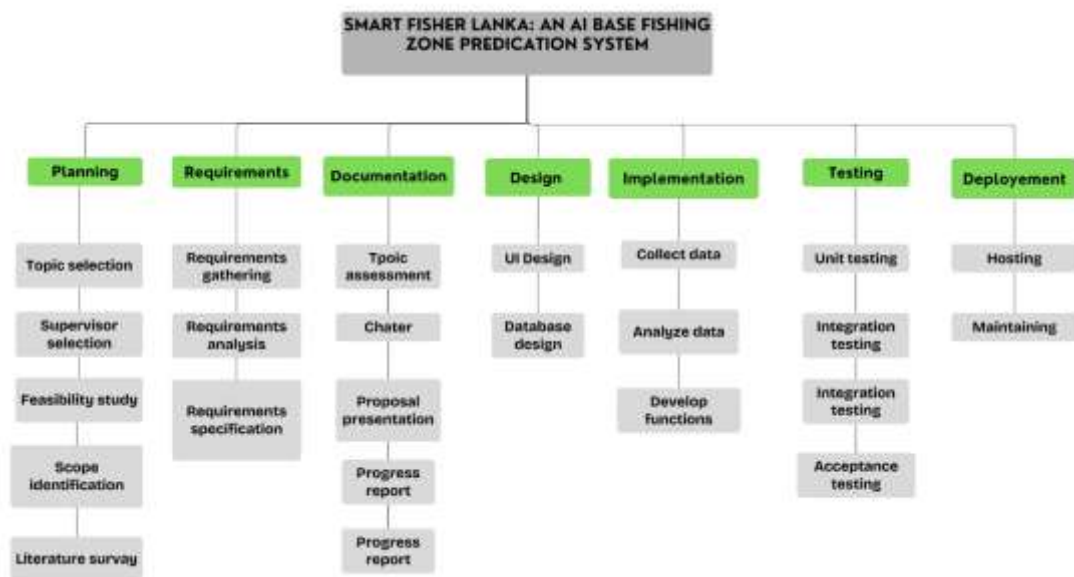


Figure 5.1: Work Breakdown Structure

## **6. DESCRIPTION OF PERSONAL AND FACILITIES**

Facilitators:

- Prof. Pradeep Abeygunawardhena– Sri Lanka Institute of Information Technology
- Mr. Ashvinda Iddamalgoda– Sri Lanka Institute of Information Technology

## 7. Budget

Table 1: Budget

| Task                                                                         | Cost                |
|------------------------------------------------------------------------------|---------------------|
| Cloud Server (model training, backend & APIs)                                | \$60 – \$200 / year |
| Domain Name Registration                                                     | \$10 – \$15 / year  |
| API Services (satellite data, maps, vision, notifications, etc.)             | \$25 (one-time)     |
| Mobile App Publishing (Google Play Developer Account)                        | \$50 – \$150        |
| Data Acquisition & Dataset Preparation (satellite, fish images, market data) | \$50 – \$150        |
| IoT Devices & Field Equipment (GPS/catch loggers, testing tools)             | \$100 – \$200       |
| Field Testing & Validation (with fishermen, market trials)                   | \$100 – \$200       |
| Contingency & Miscellaneous                                                  | \$15 – \$30         |
| Total (First Year)                                                           | \$400 – \$1000      |

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